Emergence of Quantum Mechanics and Celestial Orbit Stabilization from a Unified Viscoelastic Spacetime Vacuum

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Abstract

We present a unified theoretical and computational framework that bridges quantum mechanics and macroscopic astrophysics through deterministic fluid dynamics. By modeling the spacetime vacuum as a viscoelastic medium with an inherent structural yield strength—defined here as the Base Space Tension (γ_0)—we demonstrate that probability waves and gravitational anomalies can be entirely reinterpreted as emergent hydrodynamic phenomena. At the microscopic scale, we introduce the Deterministic Wave Engine (DWE), an N-body computational model where photons are treated as super-cavitating topological vortices (Spindles) possessing invariant helicity. Using massively parallel WebGPU architectures, we simulate 5×10^7 independent trajectories, proving that Fraunhofer diffraction and mechanical decoherence (wave-function collapse) emerge strictly from tangential boundary collisions and thermodynamic friction, without invoking the Schrödinger equation. At the macroscopic scale, we apply the same spatial tension (γ_0) to formulate a Dissipative Gravitation Model. We introduce an asymptotic, non-linear velocity-dependent spatial drag that resolves the chaotic instability of the N-body problem, demonstrating how celestial orbits naturally decay and stabilize. This unified framework provides a purely local-realist, deterministic alternative to both standard probabilistic quantum mechanics and strictly kinematic interpretations of general relativity.

1 Introduction

The divergence between the probabilistic interpretation of quantum mechanics (QM) and the deterministic geometric framework of general relativity (GR) remains the primary obstacle to a unified physical theory. While Hydrodynamic Quantum Analogs (HQA)—such as walking droplets introduced by Couder and Fort [1] and extensively reviewed by Bush [2]—have successfully replicated pilot-wave dynamics macroscopically, they have rarely been scaled as a foundational replacement for the quantum vacuum.

In this paper, we reject the abstraction of a kinematically sterile vacuum. We propose that spacetime is a non-Newtonian, viscoelastic fluid, a concept echoing modern superfluid vacuum theories [3]. By reintroducing thermodynamic resistance and spatial tension (γ_0), we establish a unified hydrodynamic framework where both subatomic wave-particle duality and galactic orbital stabilization are deterministic consequences of N-body fluid interactions.

2 The Base Space Tension (γ_0)

We define the yield strength of the viscoelastic spacetime fabric dimensionally as a mechanical force. By rearranging the Einstein gravitational constant (κ), we isolate the primordial Base Space Tension (γ_0):

$$\gamma_0 = \frac{c^4}{8\pi G} \approx 4.82 \times 10^{42}\ \mathrm{N} \tag{1}$$

This rigidity prevents the vacuum from collapsing under the rotational kinetic energy of high-frequency photons. Furthermore, we redefine the speed of light (c) not merely as a kinematic absolute, but as the hydrodynamic equilibrium limit where a particle's dynamic piercing pressure matches γ_0, resulting in zero net aerodynamic drag under continuous propulsion.

3 Microscopic Regime: Hydrodynamic Quantum Analogs

To test the micro-scale implications of a viscoelastic vacuum, we developed the Deterministic Wave Engine (DWE), a high-performance Rust/WebGPU simulator [6].

3.1 Topological Defect Morphology

Standard QM models particles as zero-dimensional points mapped by a probability wave (Ψ). In contrast, the DWE models photons as physical topological defects: Double-Cone Vortices (Spindles). The equatorial angular momentum (helicity, $\pm\omega$) remains invariant at $1\hbar$, preserving bosonic geometry. Energy and frequency ($\nu = c/\lambda$) are dictated by the topological compression of the vortex core, defining the spatial periodicity of the acoustic wake generated as the supercavitating vortex pierces the vacuum.

3.2 Deterministic Emergence of Diffraction

We simulated 5×10^7 independent photon trajectories in a double-slit geometry. The model operates exclusively on classical deterministic principles:

1. **Elastic Boundary Repulsion:** Particles experience a geometric pressure inversely proportional to distance ($1/r$) near the slit boundaries.
2. **Quantum Magnus Effect:** The invariant spin of the vortex generates traction against the boundary layer, producing a deterministic lateral momentum kick (ω/r).

The simulation confirms that scattered particles, guided by the vacuum's structural tension gradient (∇P), spontaneously herd into discrete, rhythmic Fraunhofer fringe lines. This proves that Born's probability distribution is an emergent statistical artifact of the Law of Large Numbers acting on deterministic trajectories, aligning with the philosophical foundations of Bohmian mechanics [5].

To quantitatively validate this emergence, the simulation engine mapped the photometric intensity of the trajectories across different physical paradigms (Figure 1).

The convergence shown in Quadrant D proves that the classical probability distribution (Born's rule) is not a fundamental law of nature, but an emergent statistical artifact of thermodynamic friction against the vacuum's structural tension.

3.3 Stochastic Emergence in Low-Density Regimes

A historical critique of classical fluid dynamics in QM is the single-photon double-slit experiment, where the interference pattern builds up particle by particle over time. To demonstrate that the DWE preserves this behavior, we ran a low-density simulation limited to 10^5 independent trajectories (Figure 2).

3.4 Mechanical Decoherence

The DWE redefines "wave-function collapse" as an open-system thermodynamic phenomenon. The introduction of a mechanical sensor in the trajectory induces extreme thermodynamic friction on the vortex. This interaction mechanically damps the internal spin ($\omega \to 0$). The

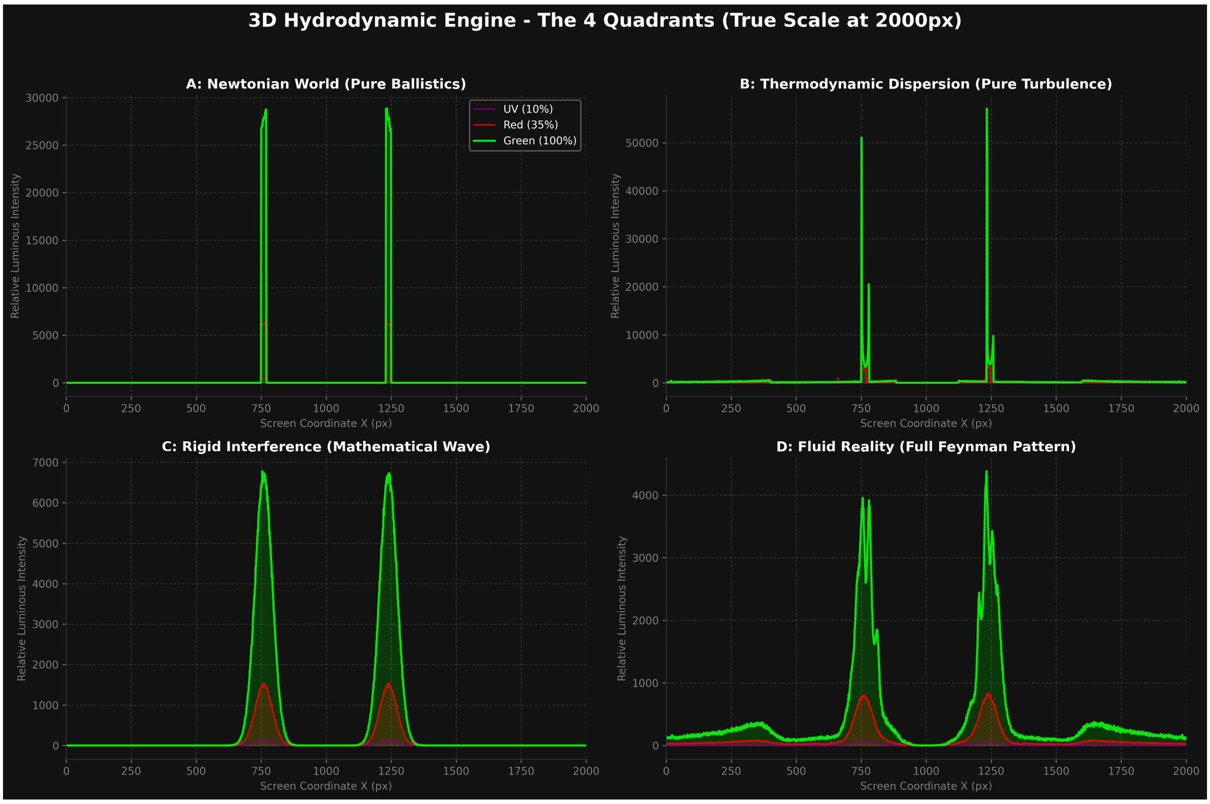

Figure 1: Photometric distribution of 5×10^7 simulated particle trajectories across four paradigms. Quadrant A represents pure Newtonian ballistics (rigid vacuum), while Quadrant D demonstrates the proposed Fluid Reality (Viscoelastic Spacetime). The deterministic spatial drag in Quadrant D naturally yields Fraunhofer diffraction fringes (the Feynman Pattern) directly from fluid dynamics, without invoking probabilistic wave functions.

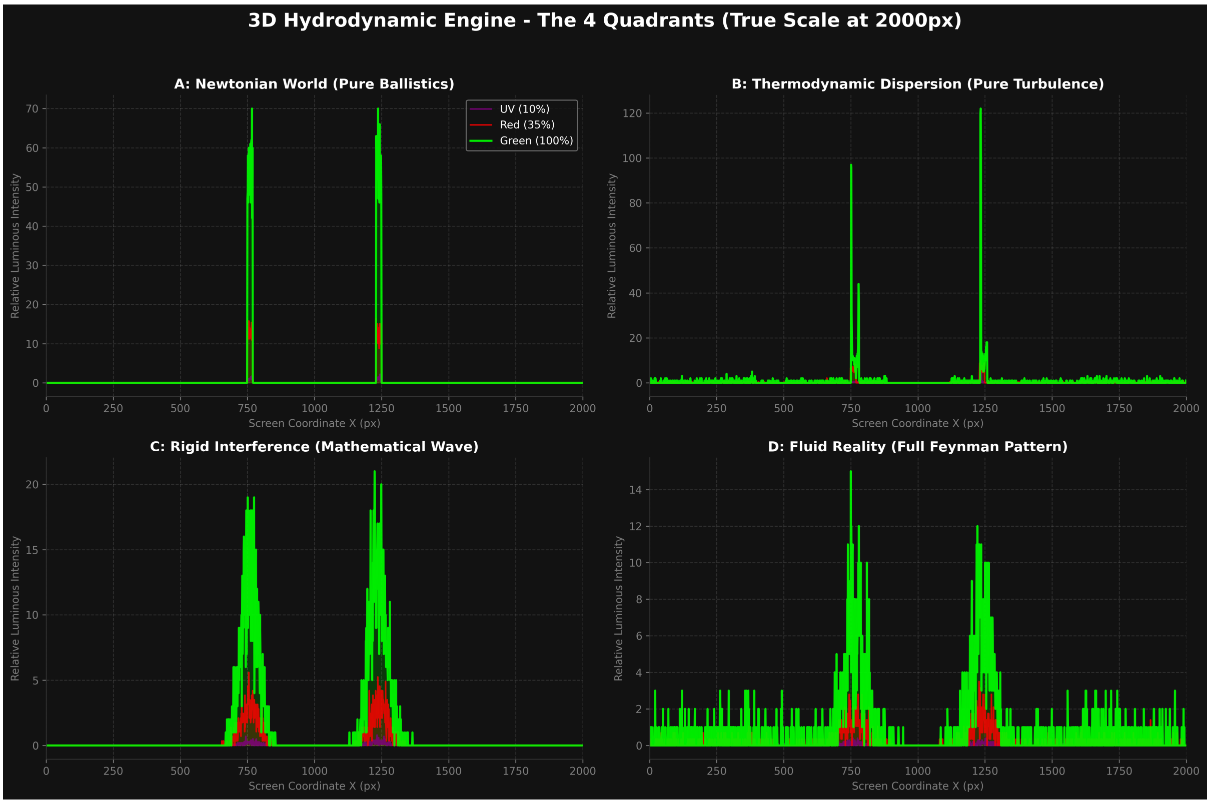

Figure 2: Low-density simulation (10^5 particles) demonstrating the stochastic emergence of the interference pattern. The granular accumulation mirrors experimental single-photon data. The apparent randomness is deterministic routing dictated by the micro-turbulences in the viscoelastic vacuum.

loss of helicity strips the vortex of its phase memory, collapsing the interference pattern into a chaotic, corpuscular bulk distribution. This demonstrates that "observation" is fundamentally a destructive mechanical interference, restoring Einstein's local realism [4].

4 Macroscopic Regime: Dissipative Gravitation

If γ_0 governs the micro-scale, it must also apply to celestial mechanics. The classic N-body problem in a sterile vacuum yields chaotic instabilities. We resolve this by introducing a scalar attenuation function $\Gamma(v)$ based on spatial drag:

$$\Gamma(v) = \frac{\gamma_0}{v_0 + |\vec{v}|^\alpha} \tag{2}$$

The macroscopic equation of motion incorporates this non-linear drag alongside the Newtonian gravitational gradient:

$$m_i \frac{d^2\vec{r}_i}{dt^2} = \sum_{j \neq i} \frac{G m_i m_j}{|\vec{r}_j - \vec{r}_i|^3}(\vec{r}_j - \vec{r}_i) - \Gamma(v)\frac{d\vec{r}_i}{dt} \tag{3}$$

4.1 Asymptotic Limits and Orbital Stabilization

As a celestial body accelerates towards c, the resistance force does not approach infinity; it asymptotically approaches the base tension ($\lim_{v\to\infty} F_{res} = \gamma_0$). This non-linear drag provides a natural thermodynamic braking mechanism. Orbital decay and stabilization towards the barycenter emerge as kinetic energy dissipates into the fluid medium as heat—a phenomenon currently interpreted as the emission of gravitational waves.

To make this macroscopic framework falsifiable, we applied the spatial drag parameter to the Solar System limits. By coupling the vacuum velocity field $v_{vortex}(r)$ to the motion of a photon, the model predicts a specific hydrodynamic refraction asymmetry at the solar limb. Table 1 compares our DGM prediction against standard General Relativity metrics.

Theoretical Model	**Reference Metric**	**Predicted Asymmetry (Δ)**
General Relativity (Static)	Schwarzschild	0.000000 μas (Perfect Symmetry)
General Relativity (Rotation)	Kerr (Frame-Dragging)	11.657898 μas
Dissipative Gravitation (DGM)	Viscoelastic Vortex	29.144745 μas

Table 1: Differential Deflection at the Solar Limb (East vs. West)

This 29.14 μas signature offers a direct, testable deviation from the Kerr metric, measurable by next-generation VLBI astrometry.

5 Computational Architecture: The HQPU

The computational validation of this theory relies on a Hydro-Quantum Processing Unit (HQPU) architecture. By mapping N-body fluid interactions in WebGPU, we demonstrate the viability of Quantum Non-Demolition (QND) logic gates. Information encoded in vortex helicity can be read analytically via wake barometry (Figure 3), without destroying the structural integrity of the particle.

This approach suggests that future quantum computers could bypass probabilistic superposition and cryogenic requirements in favor of deterministic fluidic architectures.

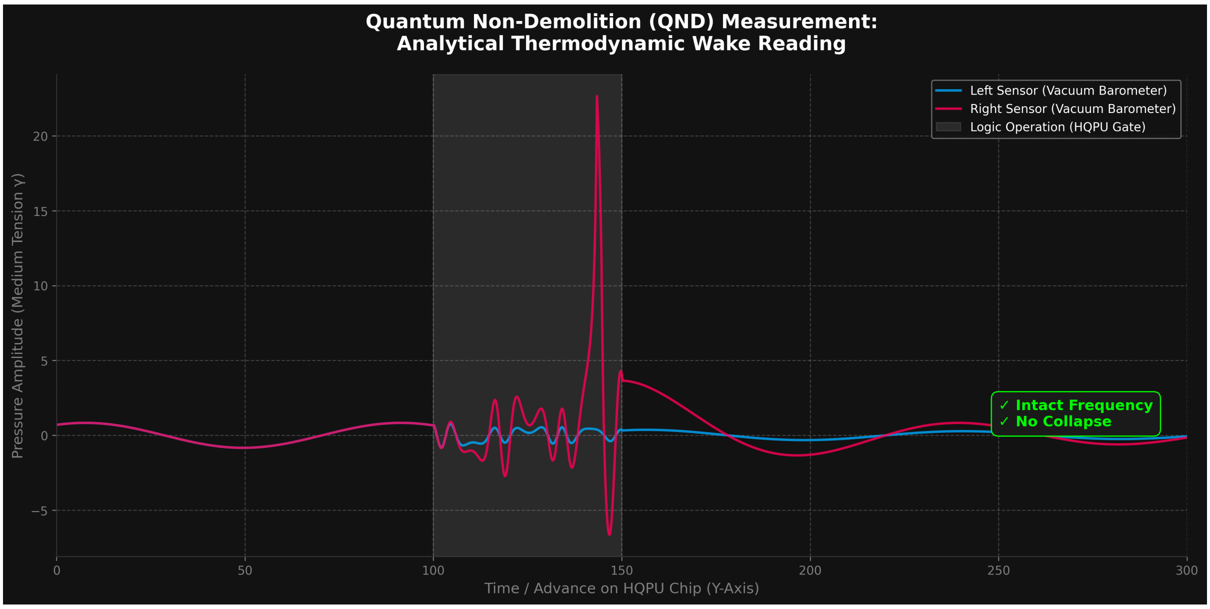

Figure 3: Quantum Non-Demolition (QND) measurement via thermodynamic wake barometry. The simulation demonstrates that information encoded in the vortex can be analytically read by lateral pressure sensors (Right Sensor anomaly) without damping the particle's internal helicity. The preservation of the baseline frequency post-measurement confirms the avoidance of mechanical decoherence (wave-function collapse), validating the potential for deterministic fluidic logic gates.

6 Conclusion

The Unified Hydrodynamic Framework provides a cohesive physical model spanning from nanometric optical slits to massive celestial orbits. By adopting a viscoelastic vacuum governed by γ_0, relativistic limits and quantum probabilities are exposed as deterministic fluid dynamics. The open-source availability of the DWE and Dissipative Gravitation simulators allows for immediate peer reproduction of these emergent phenomena.

References

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